

Orchestration Technique of Intelligent Microservices on the Edge Computing in High-Resource-Constraint Devices

Caio José Borba Vilar Guimarães

UFCG

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Outline

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- Main Objective
- Specific Objectives

Main Objective

To develop, evaluate, and validate a technique for orchestrating intelligent microservices within a distributed and collaborative Edge Computing (EC) infrastructure composed of resource-constrained devices.

- To design a clear topology architecture for the distributed and collaborative Edge Computing infrastructure;
- To implement a collaborative and distributed scheduling system that intelligently instantiate microservices on connected devices;
- To create methods and algorithms for intelligent microservices orchestration;
- To test and evaluate these solutions both in simulated environments and real IoT setups, covering scenarios typical of Smart Environments and IoT applications.

Specific Objectives

- Study paradigms of orchestration techniques and protocols for collaborative distribution among constrained devices;
Design and develop a collaborative microservices orchestration technique tailored for high-restriction devices in distributed and cooperative environments, particularly in smart environments;
- Simulate existing state-of-the-art techniques in smart environments to evaluate their performance in terms of transmission rate, energy consumption, and scalability in microservices deployment;
- Develop an infrastructure with high-restriction devices and diverse communication technologies (wired and wireless) to test selected techniques in scenarios closely resembling real-world IoT environments;

Specific Objectives

- Implement and evaluate AI microservices (Machine Learning and Deep Learning models)—assessing accuracy, inference latency, and provisioning time in both simulated and real IoT environments; Develop new methods and algorithms for orchestration and cooperative control of intelligent microservices, optimizing the use of limited resources and mitigating the limitations of existing techniques;
- Analyze and compare the performance of simulation and real-world implementations, refining techniques and protocols based on empirical data;
- Ensure the developed protocols are feasible and conformant, applying formal analysis methods to verify their viability considering system constraints like memory and processing power;
- Optimize existing algorithms and protocols by improving efficiency and reducing resource consumption, adapting based on feedback from initial implementations and tests